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Developing A Hybrid Response Algorithm for Ransomware Detection and Mitigation Using Windows Native Tools Integration

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Chapter 1

Research Introduction

1.1 Introduction

Crypto-ransomware is one of the most serious threats facing information security today. It has evolved from attacks on individuals' personal devices into systematic campaigns that directly target the file servers of companies and organizations across all sectors, including governmental, military, economic, and commercial institutions. Particularly dangerous is its use of a technique known as **in-place encryption**, which modifies the internal content of files without affecting their names or extensions, giving it the ability to bypass traditional defense systems based on file signatures. The danger is compounded by a modern trend toward **intelligent silent encryption**: a behavior that mimics the natural behavior of the system **(Benign Behavior)** by using a slow and intermittent encryption method, which allows the ransomware to spread and evade **AI/ML**-based behavioral monitoring systems for extended periods.

1.2 Problem Statement

A significant problem persists in the inability to accurately identify ransomware that employs **in-place** encryption. This issue is further complicated by the weakness of security solutions in detecting **ransomware** when encryption is performed in place, and by the shortcomings of artificial intelligence, such as the high number of false alarms and the isolation of authorized users and devices, or conversely, the misconception that silent encryption is normal behavior when it is actually a sophisticated criminal tactic. Furthermore, there is a tension between **ransomware** detection accuracy and system resource utilization, due to the heavy reliance of **AI/ML** algorithms on computational resources to filter a huge volume of alerts in order to identify the actual attack.

1.3 Purpose Statement

The purpose of this quantitative experimental study is to develop a hybrid response algorithm for the simultaneous detection and containment of ransomware, achieving a detection reliability that overcomes both the shortcomings of traditional systems based on file-fingerprinting lists and the limitations arising when the new generation of ransomware intelligently ignores **honey-files** or employs **silent** or **in-place** encryption. The study also addresses the challenges of artificial intelligence **(AI/ML)** highlighted in modern

research. The algorithm avoids probabilistic prediction based on the natural behavior of the user or system and removes the dependence on machine-learning models by establishing a real-time, non-probabilistic technical link between the suspicious **process ID** and the hardware behavioral signature extracted via the **PerfMon** tool. This is done by matching the **process ID** that attempted to access the files with the **ID** responsible for the real-time physical increase in the consumption of **CPU, RAM, and I/O** resources. Since hardware behavior represents a physical reality, it cannot be faked or mimicked programmatically by the new generation of ransomware. This behavioral match therefore detects silent encryption at the moment it occurs and enables immediate containment and isolation of the attack, which is intended to meet the needs of companies and organizations in all sectors, whether governmental, military, economic, or commercial, even if the attacker succeeds in avoiding the **honey-files** or bypassing the abnormal behavior detection of the virus. The result is a strong and inevitable defense system based on the outputs of a lightweight hybrid algorithm that combines the **FSRM** tool and the **PerfMon** tool, rather than on programmatic guesswork.

1.4 Research Questions

This study aims to answer the following research questions:

Main Research Question

1. What is the accuracy level of the proposed algorithm in addressing the probabilistic prediction flaws of AI/ML systems that rely on natural user or system behavior **(probabilistic prediction flaws)**?

Sub-questions

2. To what extent does the proposed algorithm, which correlates the PID that exceeded the threshold with the whitelists of authorized processes, contribute to reducing false alarms compared to probabilistic prediction-based systems?

3. Is there clear statistical evidence of faster containment of the expansion and spread of ransomware when using automatic file tagging technology **(FCI)** compared with traditional software isolation methods?

4. To evaluate the efficiency of the proposed algorithm in minimizing the impact on system resource utilization (CPU and RAM) compared with the heavy load imposed by AI/ML systems.

1.5 Research Objectives

The main objective of this study is to develop a lightweight hybrid algorithm that proactively detects and prevents the new generation of ransomware in real time and with deterministic reliability, isolates infected files, and terminates the malicious process, thereby overcoming the limitations of traditional signature-based systems and modern AI/ML systems. Specifically, the study aims to:

1. To evaluate the accuracy of the proposed hybrid algorithm in identifying the next generation of ransomware by linking the PID to PerfMon counters, compared with probabilistic systems that generate many false alarms.

2. To measure the difference in the speed and efficiency of automatic containment and isolation of ransomware when using the FCI and FSRM tools for tagging infected files and isolating them immediately upon detection, and to compare this automatic response with the delayed response of traditional isolation methods.

3. To test the effectiveness of reducing false-positive alarms by relying on the physical hardware signature to filter alarms, compared with the frequency of false alarms generated by probabilistic behavior in AI/ML systems.

4. To examine the system resource consumption, represented by RAM and CPU, of the lightweight hybrid algorithm based on native tools, and to measure these results against the heavy computational requirements of AI/ML-based methods.

1.6 Significance of the Study

The importance of this study lies in offering a solution to the problem of moving from theory to actual application in real work environments, which remains a major challenge in modern research, especially for large companies and government agencies that depend on probabilistic predictions produced by AI/ML.

The practical value of the proposed algorithm lies in its ability to detect crypto-in-place attacks, which deceive traditional systems by encrypting files without altering their extensions and therefore without leaving a recognizable fingerprint. The study also addresses the problem of false alarms generated by probabilistic prediction systems, which often unnecessarily isolate the devices of authorized users, causing disruptions and shutdowns of critical processes.

The study focuses on simultaneously detecting silent encryption, ensuring immediate containment and isolation of the attack, and providing a deterministic defense through an algorithm that establishes a real-time, non-probabilistic link between the suspicious process ID (PID) and the hardware behavior signature extracted via the PerfMon tool. Because hardware behavior reflects a physical reality, it cannot be faked or programmatically simulated by the new generation of ransomware.

The proposed algorithm is designed as a hybrid solution combining proactive prevention of traditional ransomware, immediate detection and prevention of next-generation ransomware, and automated containment and isolation of infected devices and files using native system tools. This design makes it a suitable solution for large corporations and organizations as well as for entities with limited budgets, owing to its resource efficiency and the absence of additional licensing costs.

1.7 Scope and Delimitations of the Study

The scope of this study is limited to companies and organizations using the Windows operating system and file-based systems, and it covers the new generation of ransomware that employs silent encryption and encryption-in-place tactics. The study focuses on measuring physical changes in hardware performance (CPU, RAM, I/O) and correlating

them with the Process Identifier (PID), and it involves developing a hybrid algorithm based on native tools. Spyware and general system-sabotage programs that do not target file encryption were excluded.

This study links malicious software behavior with its resulting hardware-level physical impact in order to gain an in-depth understanding suitable for organizations that require strong protection with minimal resource consumption. The study does not include the development of heavy or probabilistic artificial intelligence models. These boundaries, however, present an opportunity to develop a lightweight hybrid algorithm that could be extended to other operating systems.

1.8 Definition of Terms

Encryption-in-place

It is an advanced mechanism used by modern ransomware, where it modifies the file content and encrypts it in its original location on the hard drive without creating a new copy or changing the file extension, making it extremely difficult for systems that rely on ransomware signatures to detect it.

Silent Encryption

A smart encryption process that attempts to mimic permitted operations by encrypting intermittently at low consumption rates to bypass traditional monitoring systems and ensure the encryption process is completed without the user's knowledge.

Probabilistic Prediction

The methodology adopted by Artificial Intelligence and Machine Learning (AI/ML) systems for threat detection, relying on behavioral estimation based on historical patterns. Its main drawback is the margin of error, which leads to "false positives" or failure to detect previously unknown patterns.

Deterministic Defense

The approach adopted in this study, which relies on "physical certainty" rather than estimation. It establishes a direct link between a software event (opening a file) and a definitive physical impact on the hardware, eliminating false alarms and ensuring detection accuracy for the targeted behavior.

Process Identifier (PID)

A unique numerical label assigned by the operating system to each active process in the memory. In the proposed algorithm, it serves as the primary correlation factor to precisely identify the program performing encryption and link it to hardware consumption footprints.

Performance Monitor (PerfMon)

A Windows native tool used in this study to extract real-time hardware performance counters (CPU, RAM, I/O), providing a physical behavioral signature for every active process.

File Server Resource Manager (FSRM)

A feature within Windows Server systems that enables file management and classification. It is utilized in this study as a first line of defense (proactive prevention) by monitoring file access and triggering real-time alerts.

Lightweight Hybrid Algorithm

The algorithm developed in this research, which integrates native system tools to provide comprehensive protection with minimal consumption of CPU and memory resources, making it suitable for devices with limited specifications.

File Classification Infrastructure (FCI)

A built-in technology within the Windows Server system (as part of FSRM) that enables researchers to automate file classification based on content or properties. In this study, FCI is utilized as a mechanism for tagging sensitive files and applying immediate protection policies upon detecting suspicious activity from a specific Process Identifier (PID). This facilitates the "intelligent isolation" of files during an ongoing attack.

1.9 Organization of the Thesis

Chapter One establishes the general framework of the study, addressing the research problem related to the shortcomings of theoretical models in real-world applications and the significance of a deterministic solution for large-scale organizations, while defining the studys objectives, research questions, technical scope, and procedural terms such as (PID, PerfMon, FCI); moving to Chapter Two, which reviews the literature and previous studies related to advanced ransomware tactics like "Encryption-in-place," highlighting theories and models based on probabilistic prediction (AI/ML) while providing an analytical critique of prior research in a critical analytical style to illustrate the flaws of total reliance on probabilistic algorithms that cause false positives and disrupt vital operations; this paves the way for Chapter Three, which presents the research methodology and experimental design, detailing the development of the hybrid algorithm that establishes a real-time link between the suspicious Process Identifier (PID) and the hardware behavioral signature extracted via the (PerfMon) tool, along with procedures for monitoring system resource consumption; leading to Chapter Four, which presents the data analysis results and answers the research questions by providing empirical evidence of the algorithm's ability to detect silent encryption, reduce false positives generated by AI models, and achieve the equation of precise detection with minimal CPU and memory overhead; the research concludes with Chapter Five, which discusses the findings, draws conclusions regarding the effectiveness of conclusive physical evidence, demonstrates the practical implications of the automated isolation system in protecting critical operations,

and provides final recommendations for technical practice in military, governmental, and economic environments.

Chapter 2

Literature Review

This chapter reviews the literature that underpins the proposed hybrid response algorithm. It first outlines the ransomware threat model, its taxonomy, and the detection and prevention strategies documented in previous surveys [1–3]. It then examines the native Windows Server components on which the algorithm is built—the File Server Resource Manager (FSRM) with its File Classification Infrastructure (FCI), and the Performance Monitor (PerfMon)—and reviews how deception-based defenses [4, 5] and file-system behavioral detectors [6–9] have been applied against ransomware, together with honeyfile-based protection on Windows hosts [10]. The chapter concludes with an analytical comparison of ten representative studies against the proposed approach, from which the research gap addressed by this thesis is derived.

2.1 Ransomware

Ransomware is a class of malicious software that denies access to computer files, systems, or networks—most commonly by encrypting the data—and demands a ransom payment for restoring access.

Cybercriminals use ransomware to seize data and effectively hold it hostage, releasing it only upon payment. Organizations that hold sensitive data, such as personal information, financial records, and intellectual property, are the most attractive targets, since disruption of their operations increases the pressure to pay.

The threat continues to evolve as cybercriminals refine their methods of gaining access and encrypting data at speed. The accelerating digitalization of business processes, reinforced by the remote-working arrangements adopted during the COVID-19 pandemic, has placed more data in remote settings than ever before, widening the attack surface [6].

Ransomware therefore remains a significant and evolving threat in the cybersecurity landscape. As organizations strengthen their defenses, attackers continuously adapt their tactics to bypass these measures and maximize their gains, giving rise to the next-generation characteristics discussed below [2].

Types of Ransomware

- **Scareware:** uses social engineering to frighten or shock victims and manipulate them into purchasing software they do not need.
- **Screen lockers:** lock the computer screen so that it appears inaccessible. Instead of the normal desktop, the victim is shown a message demanding payment before

access is restored; the message may impersonate a law-enforcement agency and instruct the victim to send money through an online payment service.

- **Encrypting ransomware:** employs strong encryption algorithms to encrypt the data on the device and leaves a ransom note explaining the payment demanded and the steps required to regain access. Unless a recent backup exists, victims have little recourse other than meeting the attackers demands.

- **Ransomware threats:** continue to evolve and grow more severe. As new security measures emerge, attackers devise new ways to compromise the computers of individuals and enterprises. Threats such as ransomware-as-a-service (RaaS) are increasingly common: with RaaS, a buyer can purchase or rent a complete ransomware package to deploy against any target, sometimes sharing the profits with the provider.

Ransomware Detection and Response

Effective ransomware defense combines user education, continuous monitoring, and layered technology. The measures most widely recommended in the literature and by security agencies include the following:

- **Employee awareness and education:** Train employees to recognize the signs of ransomware, such as emails impersonating legitimate businesses, suspicious external links, and questionable file attachments.

- **Honeypots and honeyfiles:** A honeypot is a decoy repository of files designed to look like an attractive target; any interaction with it is inherently suspicious and can be used to detect and stop an attack in progress. Honeyfile-based defenses of this kind have demonstrated effective early detection of ransomware on Windows and Android hosts at negligible resource cost [4, 10].

- **Network and endpoint monitoring:** Continuous monitoring makes it possible to log incoming and outgoing traffic, scan files for evidence of attack (such as failed modifications), establish a baseline of acceptable user activity, and investigate deviations from it. Behavioral research similarly shows that file-access patterns—bulk encryption, renaming, and deletion—provide reliable early signals of ransomware activity [7].

- **Antivirus and anti-ransomware tools:** These solutions whitelist acceptable applications and alert administrators when a threat is detected.

Key Strategies to Prevent Ransomware Attacks

- **Update frequently**: Keeping devices updated is a simple but effective defense. Updates often include fixes that block new ransomware variants. Enable automatic updates or check regularly to avoid gaps in protection.

- **Authenticate software:** Install software only from trusted and verified sources, and enforce multi-factor authentication for stronger access control.



Figure 2.1: 8 Key Strategies to Prevent Ransomware Attacks

- **Install antivirus protection:** Antivirus software provides a first line of defense against ransomware. Email filters block malicious attachments before they reach users, and advanced firewalls add continuous monitoring and automated threat updates.
- **Whitelist software:** Whitelisting ensures that only approved programs can run on the system. Firewalls can alert security teams to suspicious applications before they connect, and new software should be blocked while a suspected breach is investigated.
- **Back up data:** Backups do not prevent ransomware, but they make recovery faster. Regular copies create a clean restore point for critical systems, which can then be wiped and restored with minimal disruption after a cyberattack.
- **Educate employees:** Human error can lead to ransomware infections. Training employees to identify phishing attempts and unsafe links is therefore crucial to overall cybersecurity.
- **Use a comprehensive security solution:** A multi-layered solution combines web filtering, AI/ML, firewalls, and email security, while sandboxing isolates applications so that suspicious behavior can be analyzed safely. Together, these measures create a strong defense against ransomware.

Next-Gen Ransomware

Ransomware attacks are no longer solely focused on encrypting files and demanding a ransom for their decryption. Modern ransomware operations exhibit a higher degree of sophistication and employ a wider range of techniques [2]:

- **Double Extortion:** Beyond encryption, attackers now frequently exfiltrate sensitive data before locking systems. This dual approach puts immense pressure on victims, as they face not only business disruption but also the threat of public data leaks if the ransom is not paid.

- **Triple Extortion:** Some ransomware groups are adding further layers of pressure, such as launching distributed denial-of-service (DDoS) attacks against the victim's infrastructure or directly contacting customers or business partners to further coerce payment.
- **Ransomware-as-a-Service (RaaS) Evolution:** The RaaS model continues to lower the barrier to entry for aspiring cybercriminals. Advanced RaaS platforms are providing affiliates with sophisticated tools, infrastructure, and support, enabling even less skilled actors to launch highly effective attacks. These platforms often incorporate advanced features like automated deployment, evasion techniques, and negotiation support.
- **Targeting Critical Infrastructure:** There is a growing trend of ransomware attacks targeting critical infrastructure sectors such as healthcare, energy, and transportation. These attacks can have severe real-world consequences, making victims more likely to pay the ransom quickly.
- **Supply Chain Attacks:** Cybercriminals are increasingly targeting organizations within a supply chain to gain access to multiple downstream victims. By compromising a single, less protected entity, attackers can potentially disrupt numerous businesses simultaneously.
- **Sophisticated Evasion Techniques:** Next-gen ransomware employs advanced techniques to evade detection by traditional security solutions. These include polymorphic malware that changes its code to avoid signature-based detection [7], living-off-the-land tactics that abuse legitimate system tools to carry out malicious activities, and the exploitation of zero-day vulnerabilities.
- **Increased Focus on Data Exfiltration:** The value of exfiltrated data extends beyond extortion itself. Cybercriminals are increasingly using stolen data for other malicious purposes, such as selling it on the dark web or leveraging it for follow-on attacks.

2.2 File Server Resource Manager (FSRM)

FSRM is a role service in Windows Server that allows you to manage and classify data stored on file servers. You can use File Server Explorer to automatically classify files, perform tasks based on these classifications, set quotas on folders, and create reports that monitor storage usage.

It applies to Windows Server 2019, Windows Server 2016, Windows Server 2012 R2, Windows Server 2012, Windows Server 2008 R2. File Server Resource Manager only supports volumes formatted with the NTFS file system; the Resilient File System (ReFS) is not supported.

File Classification in FSRM

File Classification Infrastructure (FCI) provides insight into your data by automating classification processes. It allows you to define properties (e.g., "Confidentiality") and assign values (e.g., "High") based on the content or location of the files

- **Quota management:** Limit the space that is allowed for a volume or folder. These limits can be automatically applied to new folders that are created on a volume. You can also define quota templates that can be applied to new volumes or folders.

- **File Classification Infrastructure:** Gain insight into your data by automating classification processes so that you can manage your data more effectively. You can classify files and apply policies based on this classification. Example policies include dynamic access control for restricting access to files, file encryption, and file expiration. Files can be classified automatically by using file classification rules or manually by modifying the properties of a selected file or folder.

- **File Management Tasks:** Gain the ability to apply a conditional policy or action to files based on their classification. The conditions of a file management task include the file location, the classification properties, the date the file was created, the last modified date of the file, or the last time the file was accessed. The actions that a file management task can take include the ability to expire files, encrypt files, or run a custom command.

- **File screening management:** Control the types of files that the user can store on a file server. You can limit the extension that can be stored on your shared files. For example, you can create a file screen that doesn't allow files with an MP3 extension to be stored in personal shared folders on a file server.

- **Storage reports:** Use these reports to help you identify trends in disk usage and how your data is classified. You can also monitor a selected group of users for attempts to save unauthorized files.

File screening management for Ransomware Mitigation

Ransomware attacks against file servers continue to be reported on a regular basis. While commercial products and services exist to detect and contain ransomware, Microsoft's File Server Resource Manager (FSRM) can be leveraged as an additional, native layer of defense that prevents the encryption of files on network shares. File screening management enables administrators to create active file screens that block file extensions known to be associated with ransomware (e.g., .crypter, .locky, .wncry) and to deploy canary files—decoy files placed at the top of a directory's alphabetical order—whose modification is inherently malicious. When FSRM detects the modification of a canary file or the creation of a screened extension, it triggers an immediate alert to which an automated response can be attached. A closely related deception-based strategy, validated on Windows platforms through the R-Locker tool, is examined among the previous studies later in this chapter [10].

File Management Tasks

File Management Tasks are automated routines that run on a schedule or are triggered by specific conditions. They scan files on a server and, if the files meet specific criteria (such as a certain File Classification property or a period of inactivity), they perform a predefined action.

File Management Tasks (FMT) enable the automation of defensive responses against ransomware-classified data by allowing administrators to trigger multi-layered actions, such as enforcing read-only attributes to halt encryption, isolating active user sessions to contain the threat, generating real-time forensic reports, and dispatching instant administrative alerts the moment suspicious activity is detected.

2.3 Performance Monitor (Perfmon)

PerfMon is a built-in performance monitoring tool for Windows systems. It provides a wide range of performance statistics and graphs, monitoring activities such as CPU and memory usage on local or remote computers, and can be used to diagnose the performance of hardware, software services, and applications. The tool supports configurable alert thresholds with automatic actions, report generation, and the review of historical performance data.

Within this study, Performance Monitor (PerfMon) is treated as a native Windows analytical sensor: a high-fidelity telemetry source that instruments system resources in real time through performance counters, capturing low-level activities such as disk I/O frequency, CPU cycles per process, and memory cache faults. These counters supply the data streams from which the hybrid response algorithm derives the hardware behavioral signature of each process and triggers automated mitigation when a process deviates from its established baseline—in effect, identifying the cryptographic footprint of malicious encryption before it completes.

2.4 Case studies / Previous Studies

To establish a robust technical foundation and to position the proposed algorithm with respect to the state of the art, this section reviews seminal studies and previous research efforts that addressed ransomware detection and prevention, focusing on their methodologies and the technical gaps they aimed to resolve. Five case studies are examined in detail, followed by five foundational studies on behavioral and deception-based detection; the chapter then closes with a consolidated comparison of all ten works against the proposed approach.

First Case Study [1]

- **Reference Study Characterization**
  The study by Herrera Silva et al. (2019) is a comprehensive survey that established the theoretical foundations for Situational Awareness (SA) in the context of ransomware attacks. The authors employed an analytical methodology to survey a wide range of scientific literature, resulting in a precise taxonomy of detection and prevention parameters across various cognitive levels. The study maps ransomware defense onto the five phases of situational awareness—perception, comprehension, projection, decision, and action—and links the monitored parameters to the digital footprints left by the malware at both the operating system and network levels. However, the limitations of the study were evident in its purely taxonomic and normative approach. The models summarized failed to provide genuinely proactive

solutions, as most proposed detection mechanisms only operate post-infection, after the ransomware has successfully embedded itself and commenced its destructive activity. This renders them "reactive" rather than "preventative" systems. Furthermore, a significant deficiency was noted in immediate isolation and response mechanisms; the surveyed studies lacked the capability to autonomously suppress an attack in its initial moments. Additionally, an over-reliance on personal computing and Android environments was observed, leaving a critical security gap in solutions tailored for enterprise infrastructure, specifically Windows Server environments

- **Comparative Analysis**
  The current research aligns with Herrera Silva et al. (2019) on several fundamental pillars, most notably the adoption of situational awareness as a core defensive strategy. Nevertheless, this study diverges in its procedural orientation: where the reference study is limited to passive monitoring and post-infection analysis, this research transitions toward proactive active defense. A further distinction lies in the application environment, as this research is specifically engineered for Windows Server protection, thereby addressing the automated response deficiencies evident in the surveyed literature.

- **Addressing the Gaps: Methodological and Practical Innovations**
  This research seeks to mitigate the methodological and practical shortcomings identified in the Herrera Silva et al. (2019) survey, specifically delayed response and weak proactivity, through the following technical innovations:

  1. Mitigating Proactivity Deficiencies: Rather than awaiting the malwares execution and the subsequent onset of encryption, the File Server Resource Manager (FSRM) was deployed as a Zero-Day Prevention layer. This layer proactively blocks the creation of suspicious file extensions at the initial attempt of compromise.
  2. Addressing the Isolation Gap: This study resolved the "failure of immediate response" by programming an Autonomous Isolation system. This mechanism intercepts the ransomware's progression the moment a Honeyfile is accessed, eliminating the latency associated with human intervention or prolonged cloud-based analysis.
  3. Achieving Hardware Certainty: To counter the dilemma of False Positives, the Performance Monitor (PerfMon) was integrated as an ultimate safety valve. By monitoring the inevitable physical impact of encryption on hardware resources (CPU/Disk I/O), the system ensures attack detection even if the malware successfully bypasses initial software-based traps and obfuscation layers.

- **Literature Gaps vs. Implementation Results**

Technical Pillar	Herrera Silva et al. (2019)	Scientific Gap	Proposed Solution	Implementation
Nature of Work	Survey and taxonomy of parameters.	No applied model for integrated parameters.	Multi-layered Defense architecture.	Hybrid system: FSRM, Honeyfiles, PerfMon.
Target Env.	Individual users and Android.	Insufficient focus on enterprise servers.	Customizing defense for servers.	Windows Server lab setup.
Prevention	Post-infection detection.	No "Moment Zero" prevention tools.	Proactive extension blocking.	FSRM malicious extension blocking.
Accuracy	Behavioral (high error risk).	High False Positives and latency.	Honeyfiles as certainty indicator.	100% certainty upon access.
Response	Manual "Rapid Recognition".	Gap between detection and action.	Automated Immediate Isolation.	Automated trigger to kill processes.
Safety Valve	Theoretical hardware monitoring.	Malware may bypass software traps.	Monitoring physical hardware impact.	PerfMon-based encryption detection.

Table 2.2: Final Comparison: Literature Gap vs. Proposed Hybrid System

Second Case Study [2]

The study by Moussaileb et al. (2021) [2], titled “A Survey on Windows-based Ransomware Taxonomy and Detection Mechanisms: Case Closed?”, is a comprehensive systematic review that organizes the ransomware attack lifecycle into four stages (delivery, deployment, destruction, and dealing) and provides a strategic roadmap for future research. Its scope, however, is restricted to theoretical classification, and the survey itself acknowledges the difficulty of achieving autonomous early detection before the destruction phase. The present research addresses precisely this gap by implementing a hybrid applied framework of FSRM and PerfMon that ensures deterministic certainty and an immediate “kill process” response during the deployment phase, offering a practical solution that moves beyond the latency and limitations of the detection mechanisms surveyed in that work.

Comparison Aspect	Moussaileb et al. (2021) Study	Proposed Hybrid Solution (FSRM + PerfMon)
Nature of Work	Theoretical survey and literature classification.	Practical and implemented framework.
Intervention Phase	Monitors solutions across all stages (mostly Phase 3: Destruction).	Focuses on neutralizing the attack during Phase 2 (Deployment).
Prevention Mechanism	Reviews existing mechanisms like Backups and ACLs.	Immediate and forced prevention via FSRM automation scripts.

Comparison Aspect	Moussaileb et al. (2021) Study	Proposed Hybrid Solution (FSRM + PerfMon)
Detection Accuracy	Relies on ML and API behavior (susceptible to evasion/deception).	Relies on Physical Certainty (PerfMon) (non-evadable/immune to deception).

Table 2.3: Comparative Analysis: Theoretical Surveys vs. Proposed Practical Hybrid Framework

Third Case Study [3]

The study by Urooj et al. (2022) [3], titled “Ransomware Detection Using the Dynamic Analysis and Machine Learning: A Survey and Research Directions”, establishes the importance of dynamic analysis as a practical alternative to static systems, which are ineffective against obfuscation techniques. Despite its comprehensiveness, the survey also exposes a resource dilemma in machine-learning research—the trade-off between detection accuracy and system efficiency—which can degrade server performance and delay real-time response. The present research addresses these limitations through a lightweight dynamic-analysis methodology that replaces probabilistic models with the principle of deterministic certainty, using the native system tool FSRM together with performance counters from PerfMon to achieve proactive prevention, instantaneous response, and detection grounded in the certainty of honeyfile access, thereby providing a practical server-side solution that avoids the computational burden and false positives identified in that survey."

Comparison Pillar	Urooj et al. (2022) Study	Proposed Solution (FSRM + PerfMon)
Methodology	Relies on "Probabilistic" Machine Learning (ML) models.	Relies on "Deterministic and Physical Certainty."
Resource Consumption	High overhead due to complex AI algorithm processing.	Lightweight; utilizes native Windows tools and performance counters.
Detection Accuracy	Prone to False Positives (FPs) due to behavioral estimation.	Absolute accuracy; touching a Honeypot or using a blacklisted extension is definitive proof.
Response Speed	High latency; requires significant time for data analysis.	Instantaneous; detection triggers immediate "Kill Process" via FSRM scripts.
Countering Stealth Encryption	Relies on API analysis (which can be obfuscated or deceived).	Relies on PerfMon to monitor hardware impact (unforgeable physical footprint).

Comparison Pillar	Urooj et al. (2022) Study	Proposed Solution (FSRM + PerfMon)

Table 2.4: Comparative Analysis: ML-based Approaches (Urooj et al.) vs. Proposed Hybrid Deterministic Framework

Fourth Case Study [4]

The study by Gómez-Hernández and García-Teodoro (2024) [4], titled "Lightweight Crypto-Ransomware Detection in Android Based on Reactive Honeyfile Monitoring", developed an experimental detection system for Android that uses the FileObserver service to monitor honeyfiles reactively, achieving high detection accuracy at near-zero resource consumption—an efficiency goal closely aligned with that of the present research. The present work nevertheless bridges the gap left by that study by shifting the focus to enterprise-grade environments (Windows Server) and by introducing a multi-layered defense that overcomes the limitations of standalone traps. The layered design comprises a proactive layer that uses FSRM to block known extensions, honeyfiles as a primary line of defense that enforce immediate isolation upon interaction without any need for analysis, and PerfMon as a final fail-safe that provides physical certainty by monitoring the inevitable hardware impact of encryption, thereby ensuring the detection of sophisticated attacks that might otherwise bypass deceptive files.

Technical Pillar	Previous Android 2024	Scientific Gap	Logical Solution	Research Implementation
Operational Env.	Experimental system for Android Mobile OS.	Vulnerability of Windows Servers to high-cost attacks.	Transitioning defense to enterprise servers.	Full deployment on Windows Server.
Detection Method	Single-layer via "FileObserver" Honeyfiles.	System can be bypassed if malware avoids baits.	Multi-dimensional "Hardware-Software" verification.	PerfMon tracking alongside software traps.
Prevention Strategy	Reactive detection after infection starts.	Absence of "Moment Zero" blocking tools.	Activating proactive extension-filtering.	FSRM as a "Gatekeeper" to block files.
Response Latency	Focused on reporting unauthorized access.	High latency between detection and action.	Automation via an autonomous "Kill-Switch."	Automated Trigger for "Immediate Isolation."

Technical Pillar	Previous Android 2024	Scientific Gap	Logical Solution	Research Implementation
Certainty & Reliability	Relies on software triggers (False Positives).	Software hooks can be masked by ransomware.	Utilizing "Hardware Certainty" safety valve.	PerfMon fail-safe for "Stealth Encryption."
Computational Cost	Lightweight (Avoids heavy ML/AI consumption).	Efficiency achieved at the cost of overall defense depth.	Balancing "Lightweight" with "Defense-in-Depth."	Layered defense with minimal CPU overhead.

Table 2.5: Comparative Analysis: Previous Android-based Research vs. Proposed Enterprise Hybrid Framework

Last Case Study [5]

Muntaha et al. (2025) [5] examined the design of adaptive honeypots for advanced cybersecurity threat detection. The study reviews low-, medium-, and high-interaction honeypot categories and hybrid models, evaluates tools such as Honeyd, Dionaea, Kippo, and Honeypot-as-a-Service in cloud, industrial, and IoT ecosystems, and discusses the integration of AI-driven components—such as ChatGPT-based attacker engagement and containerized deployments using Kubernetes—to strengthen resilience against advanced persistent threats and zero-day attacks, with ransomware detection presented among the real-world use cases. As a conceptual framework, however, the work stops short of an implemented detection pipeline, and the high-interaction, AI-supported architectures it surveys depend on dedicated infrastructure and continuous data analysis, which can introduce resource overhead and response latency during rapid-onset attacks such as high-speed ransomware encryption.

To address these challenges, the present research proposes a hybrid response algorithm that shifts the focus from external AI models to deep integration with Windows native tools (e.g., FSRM, FCI, PowerShell, and PerfMon telemetry). By relying on native OS-level instrumentation, the proposed system minimizes external dependencies, reduces resource consumption, and ensures near-instantaneous automated mitigation and isolation of compromised processes, transforming the defense strategy from a reactive detection-only model into a proactive response-and-mitigation model.

Feature	Previous Research (AI-Based)	Current Research (Hybrid Native)
Primary Focus	High-fidelity detection & forensic acquisition.	Instantaneous response & mitigation.
Technology Stack	AI/ML models & External AI Inference.	Windows Native Tools (PowerShell, WMI, FSRM).

Feature	Previous Research (AI-Based)	Current Research (Hybrid Native)
Resource Impact	Heavy computational overhead (High CPU/RAM).	Minimal resource footprint (Lightweight).
Mitigation Speed	Latency during large-scale dataset processing.	Near-zero latency via native OS hooks.
System Dependency	Relies on external models and large data.	Zero external dependencies; utilizes built-in OS tools.
Strategic Goal	Reactive 'Detection-only'.	Proactive 'Response-and-Mitigation'.

Table 2.6: Comparative Overview: AI-Driven Detection vs. Proposed Hybrid Native Response Framework

Additional Foundational Studies

Five further foundational studies directly shaped the design of the proposed algorithm, particularly its rationale for linking file-system deception with hardware-level verification [6–10].

Kharraz et al. [6] analyzed 1,359 ransomware samples belonging to 15 families collected between 2006 and 2014, demonstrating that abnormal file-system activity—large volumes of I/O requests and manipulation of the NTFS Master File Table (MFT)—offers a reliable basis for detecting even zero-day ransomware. This measurement study established the file-system behavioral footprint on which the present research builds; however, it stopped short of an autonomous containment architecture, and its threat model predates the modern silent and in-place encryption techniques addressed in this thesis.

Scaife et al. [7] developed CryptoDrop, an early-warning detector that combines three behavioral indicators: file-type change, similarity between the content read from and written to a file, and the volume of file deletions. CryptoDrop detected 100% of the 492 evaluated samples with a median loss of only ten files. Its response, however, depends on a user being present to confirm the alert and suspend the malicious process, and the detector requires dedicated instrumentation of the host.

Continella et al. [8] proposed ShieldFS, a kernel driver that maintains adaptive models of benign file-system activity and transparently rolls back malicious writes through a copy-on-write mechanism. ShieldFS achieved 97.7% detection on 305 unseen samples while keeping all protected files recoverable. The cost of this capability is a custom kernel driver and a training period, and its statistical models remain exposed to misclassification—constraints that are difficult to justify in production server environments.

Kharraz and Kirda [9] presented Redemption, which buffers all storage I/O transparently and monitors per-process I/O request patterns, guaranteeing zero data loss for the evaluated ransomware families at an average overhead of 2.6%. Redemption nonetheless requires modification of the operating system kernel, which conflicts with the requirement—central to this thesis—of a solution assembled exclusively from native, unmodified platform components.

Finally, Gómez-Hernández et al. [10] introduced R-Locker, a tool for Windows platforms that deploys honeyfiles and automatically triggers countermeasures the moment one of them is touched, achieving effective early detection at negligible resource cost. R-Locker validates the deception paradigm adopted in this study, yet it remains a single detection layer: an attacker who identifies and avoids the traps can proceed unimpeded. This limitation directly motivates the multi-layered design of the proposed algorithm, in which deception-based alerts are cross-verified against hardware-level encryption signatures.

2.5 Comparative Analysis of Previous Studies and the Proposed Approach

To consolidate the review, Table 2.7 positions the ten studies discussed in this chapter against the proposed hybrid response algorithm. The comparison covers the platform and scope of each work, its detection or prevention technique, its reported outcome, its principal limitation, and the corresponding differentiator of the proposed approach.

Table 2.7: Comprehensive Comparison of Ten Previous Studies and the Proposed Hybrid Response Algorithm.

Study	**Platform / Scope**	**Detection or Prevention Technique**	**Reported Outcome**	**Principal Limitation**	**Comparison with the Proposed Approach**
Herrera Silva et al. (2019) [1]	Survey; multi-platform	Situational-awareness taxonomy of detection and prevention parameters, mapped to five SA phases (perception, comprehension, projection, decision, action).	Consolidated early-detection parameters and a research agenda; no detection system implemented.	Conceptual contribution; reviewed detection is largely post-infection; no autonomous response for enterprise servers.	Operationalizes situational awareness as a working Windows Server system that fuses FSRM alerts with PerfMon counters and executes automated containment.
Moussaileb et al. (2021) [2]	Survey; Windows ransomware	Systematic taxonomy of the ransomware lifecycle (delivery, deployment, destruction, dealing) and review of detection mechanisms.	Structured research roadmap for detection and prevention research.	Theoretical classification; acknowledges the difficulty of reliable early autonomous detection before the destruction phase.	Turns the lifecycle taxonomy into practice: native tools neutralize the attack during the deployment phase through deterministic PID verification.
Urooj et al. (2022) [3]	Survey; dynamic analysis with ML, multi-platform	Review of dynamic-analysis pipelines that use machine learning and deep learning for ransomware detection.	Comprehensive synthesis of features, datasets, and algorithms, with research directions.	ML pipelines are computationally heavy and probabilistic, with a false-positive risk and real-time response delays.	Replaces statistical inference with a deterministic PID–hardware correlation running at near-zero overhead on native tools.
Gómez-Hernández & García-Teodoro (2024) [4]	Android; end-user devices	Reactive monitoring of deployed honeyfiles using the FileObserver (inotify) service.	High detection accuracy with almost zero resource consumption.	Mobile scope; single deception layer that fails if traps are avoided; detection-oriented response.	Extends the honeyfile concept to enterprise Windows Server defense, adding FSRM extension screening, a PerfMon hardware fail-safe, and automated isolation.
Muntaha et al. (2025) [5]	Honeypot frameworks (review)	Review of adaptive honeypots: low-, medium-, and high-interaction models, HaaS, AI-driven attacker engagement, and containerized deployment.	Design guidelines for honeypot-based threat detection, including ransomware use cases.	Conceptual framework; high-interaction and AI-enhanced honeypots require dedicated infrastructure and analysis latency.	Uses deception inside the production OS with a minimal footprint and PID-keyed automated response, without external decoy infrastructure.

(continued on next page)

Study	Platform / Scope	Detection or Prevention Technique	Reported Outcome	Principal Limitation	Comparison with the Proposed Approach
Kharraz et al. (2015) [6]	Windows; measurement study	Large-scale analysis of 1,359 samples from 15 families; abnormal file-system activity, I/O requests, and MFT protection.	Demonstrated that zero-day ransomware is detectable through file-system activity monitoring.	Prototype-era study; predates silent and in-place encryption; no automated enterprise containment.	Builds on the file-system insight while adding real-time hardware telemetry and an autonomous kill-and-isolate response.
Scaife et al. (2016) [7]	Windows; end-host detector (CryptoDrop)	Three behavioral indicators: file-type change, read/write similarity, and deletion volume; user-mediated suspension.	100% detection of 492 samples; median loss of ten files.	Custom detector instrumentation; response depends on user confirmation to suspend the process.	Requires no custom sensor stack: FSRM, FCI, and PerfMon perform the sensing, and the response is fully automated.
Continella et al. (2016) [8]	Windows; kernel driver (ShieldFS)	Adaptive models of benign file-system activity with copy-on-write rollback of malicious writes.	97.7% detection on 305 unseen samples; protected files always recoverable.	Requires kernel-driver development and a training period; statistical models can misclassify.	Avoids kernel modifications entirely; the evidence chain from PID to hardware counters is deterministic rather than model-based.
Kharraz & Kirda (2017) [9]	Windows; kernel filter (Redemption)	Transparent buffering of all storage I/O with per-process I/O pattern analysis.	Zero data loss for the evaluated families; average overhead of 2.6%.	Requires modification of the operating system; detection is behavior-model based.	Uses only out-of-the-box native components, oriented toward enterprise file servers with no custom kernel code.
Gómez-Hernández et al. (2018) [10]	Windows; honeyfile tool (R-Locker)	Honeyfile deployment with automatic countermeasures upon trap access.	Effective early detection at negligible resource cost.	Standalone deception layer with no secondary verification if traps are avoided.	Multi-layer design in which honeyfile alerts are cross-verified against hardware-level encryption signatures.

Three observations emerge from Table 2.7. First, the survey studies [1–3] define the problem space and its monitored parameters, but they leave the deployment question open. Second, the behavioral detectors [6–9] demonstrate that file-system activity betrays ransomware early; however, each depends on custom instrumentation or kernel modification and delegates the final response to a user or to a statistical model. Third, the deception-oriented works [4, 5, 10] validate the lightweight trap paradigm but remain single-layer defenses that can be bypassed once the traps are identified. The proposed hybrid response algorithm addresses the intersection of these gaps: it retains the early-warning strength of deception and hardware telemetry while eliminating custom components, and it closes the detection-to-response loop through deterministic, PID-keyed automated isolation built entirely from native Windows Server tools.

Chapter 3

Research Methodology

3.1 Introduction

This chapter delineates the scientific methodology and the experimental procedures undertaken to develop and evaluate the effectiveness of the proposed Hybrid Response Algorithm in detecting and mitigating next-generation ransomware. The primary objective of this research is to shift from the probabilistic prediction-based defense modelsprevalent in AI and Machine Learning-driven systems, which often suffer from high false-positive ratestowards a model grounded in proactive, real-time "Physical Certainty." This is achieved through a direct correlation between the behavioral patterns of ransomware (specifically silent and in-place encryption) and the resulting physical hardware resource consumption.

Furthermore, this chapter provides a detailed account of the research's conceptual framework and the quantitative experimental design. It describes the isolated tested environment utilized for implementing and testing the algorithm's response. Additionally, it explores the integration of native Windows Server tools, such as File Server Resource Manager (FSRM) and Performance Monitor (PerfMon), and how they are orchestrated within a unified hybrid response framework. This integration aims to detect stealthy encryption tactics while ensuring an immediate, deterministic response with minimal system overhead and near-zero false alarms.

Finally, the chapter covers the independent and dependent variables subjected to measurement and discusses the evaluation criteriaboth statistical and technicalused to assess the algorithm's accuracy and speed in comparison to traditional defenses and AI/ML-based systems.

3.2 Conceptual Framework

This section reviews the theoretical and methodological framework upon which the study is based, linking the nature of the cyber threat (ransomware) with the proposed technical solution (hybrid response algorithm using native Windows tools), explaining the philosophy of the experimental design and the logical relationships between the research variables.

3.2.1 Research Conceptualization

The conceptual framework of this study stems from addressing a clear security and methodological gap in current ransomware defense systems. Traditional solutions rely either on extension detection (which fails against silent, in-place encryption) or on probabilistic AI models (which consume significant resources and generate false positives). The study's conceptual framework is based on the concept of "physical certainty," which shifts cybersecurity defense from probabilistic prediction to deterministic physical verification by linking two defensive layers:

- **The proactive behavioral layer:** This involves monitoring early attack indicators through file system surveillance and honeypot activity.
- **The physical hardware layer:** This involves measuring the deterministic physical impact of encryption on the processor and hard drive.

3.2.2 Material Certainty versus Probabilistic Prediction

The methodology is based on the premise that ransomware, no matter how sophisticated its silent encryption techniques, cannot conceal its impact at the hardware layer during the intensive encryption process.

- **Inputs**: These are the technical indicators from PerfMon (e.g., CPU Usage, Disk I/O Write) and FSRM (e.g., Honey-files access).
- **Processing**: The hybrid response algorithm that links the process ID (PID) to the actual resource consumption.
- **Outputs**: The deterministic response decision, which involves isolating the process and tagging the files via FCI.

3.2.3 Controlled Laboratory Experimental Design

This study adopts a Controlled Laboratory Experimental Design as its primary methodological framework to test hypotheses and quantitatively evaluate the performance of the proposed algorithm. The choice of an isolated virtual laboratory setup is dictated by three distinct methodological and technical rationales:

- **Control of Extraneous Variables (System Noise Control):** This methodology ensures the suspension of all volatile operating system background services and random background tasks, effectively freezing the environment in a uniform idle state known as the Baseline Simulation. This procedural control guarantees that any measured spike in the dependent variables (such as hardware resource consumption and response latency) is deterministically and exclusively attributed to the ransomware attack and the execution of the proposed algorithm, rather than systemic background noise.
- **Absolute Safety and Threat Containment:** Given that the experimental phase involves handling live, destructive "in-the-wild" ransomware samples (such as WannaCry), the isolated testbed guarantees secure containment via a private host-only virtual network configuration ($Host - Only$). This architectural design completely

eliminates the risk of any lateral movement, propagation of the threat, or data leakage beyond the boundaries of the experimental laboratory environment.

- **Internal Validity and Reproducibility:** The experimental design relies on rigid system state management using immutable virtual machine restoration anchors (CleanăOperatingăStateăSnapshots). This configuration allows for consecutive injection iterations and stress tests to be repeatedly initiated from the exact same temporal baseline and environmental conditions with microsecond accuracy, thereby ensuring statistical stability, data consistency, and high internal validity during the evaluation phase.

3.2.4 Research Variables Framework

In accordance with the requirements of quantitative experimental research, the conceptual framework defines the measurement areas through a network of relationships between the following variables:

- Independent Variables:Ransomware Encryption Technique: This includes traditional extension-changing encryption and silent/in-place encryption without name changes.
- Dependent Variables (Performance Parameters):
  - Response Latency: Measured in milliseconds (ms).
  - Resource Footprint: Measured via processor time (Processor Time) and disk density (Disk Bytes/sec)
  - Detection Accuracy: Statistically measured via the Confusion Matrix (the ratio of true to false positives).

3.2.5 Conceptual Framework Diagram

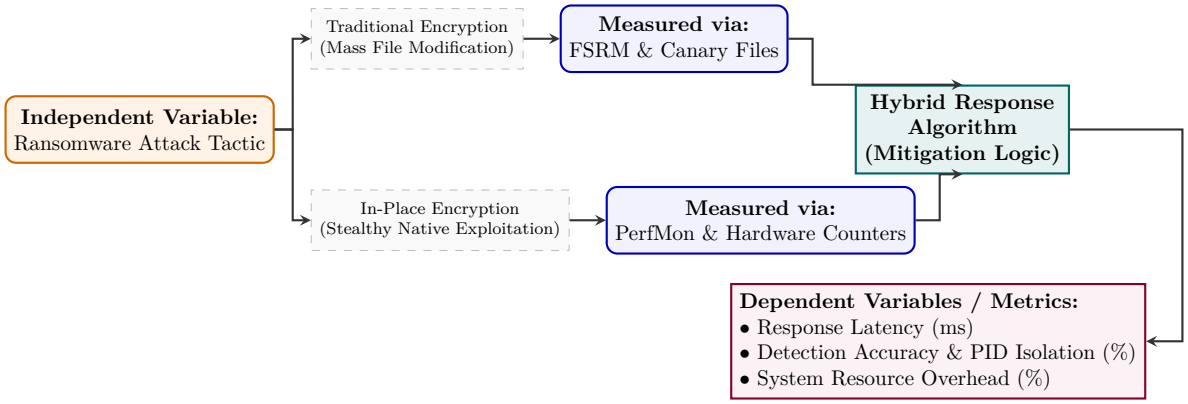


Figure 3.1: Conceptual Framework of the Hybrid Response Algorithm and Experimental Variables.

3.2.6 Conceptual Framework and Variables Alignment Matrix

This matrix establishes the logical alignment between the empirical research questions, the conceptualized variables, and the specific native system instruments used to quantify performance within the isolated testbed:

No.	Research Question / Objective	Independent Variable(s)	Dependent Variable(s)	Measurement Instrument & Unit
1	**Assessing Proactive Detection Rate:** How effectively can the algorithm detect known ransomware tactics?	**Ransomware Attack Type** (Categorical: Standard Extension-changing vs. Idle State)	**Detection Accuracy** (Quantitative: True Positives / False Positives)	`FSRM File Screening` & `Honeypots Event Logs` *(Unit: Alert Count / Percentage %)*
2	**Evaluating Physical Certainty:** Can the algorithm identify stealthy in-place encryption via hardware footprints?	**Encryption Technique** (Categorical: Silent In-place Modification)	**Hardware Resource Consumption** (Quantitative: CPU & Disk Overhead)	`PerfMon Counters` *(Units: % Processor Time, Disk Bytes/sec)*
3	**Measuring Mitigation Latency:** What is the time latency required to isolate a malicious PID after detection?	**Hybrid Trigger Integration** (Deterministic binding of FSRM and PerfMon via PowerShell)	**Response Latency (Time-to-Mitigate)** (Quantitative: Execution speed)	`PowerShell Stop-Process Execution Timestamp` *(Unit: Milliseconds - ms)*
4	**Ensuring System Stability:** How does the native integration affect the server's baseline performance?	**Defensive Architecture** (Native integration vs. Baseline Idle State)	**Resource Footprint (System Overhead)** (Quantitative: Impact on target server)	`PerfMon Baseline Verification Logging` *(Units: % CPU Load, Memory Footprint)*

Table 3.1: Variables and Conceptual Alignment Matrix

As mapped in Table 3.1, the conceptual alignment bridges the theoretical security gaps with measurable empirical metrics. Each specific objective corresponds directly to a technical dependent variable captured by the integrated native system monitors. This structured approach ensures that the subsequent data analysis layer provides statistical validation for the hybrid response algorithm's performance under various stress test scenarios.

3.2.7 Research Design and Methodological Decision Card

The following table documents the core methodological decisions regarding the research framework, experimental environment, and technical rationale to ensure the internal va-

lidity and reproducibility of the study:

No.	Methodological Decision Item	Adopted Choice in the Study	Methodological & Technical Rationale
1	Nature of the Study	Quantitative Experimental Laboratory Research	The study aims to measure precise digital and physical performance metrics (e.g., response latency and hardware consumption) and practically evaluate a mitigation algorithm.
2	Research Design	Controlled Laboratory Experimental Design	This is the most appropriate design for cybersecurity research requiring algorithm testing in rigorous, protected environments to guarantee internal validity and isolation.
3	Primary Data Collection Method	Primary Automated Data Logging & Direct Observation	The study relies completely on primary, real-time data generated directly by native operating system sensors upon injecting malware attacks.
4	Experimental Setting	Fully Isolated Virtual Testbed	To ensure absolute cybersecurity isolation, preventing any lateral threat leakage or escape of active ransomware samples (e.g., WannaCry, Ryuk) to external production networks.
5	Control and Configuration Type	Strict Control of Extraneous Variables (Baseline Simulation)	To neutralize system background noise, ensuring that any variance in the performance metrics is strictly attributed to the impact of the proposed hybrid algorithm.
6	Unit of Analysis and Observation	Digital Processes and Performance Counters (PID & Hardware Counters)	To enable the algorithm to execute deterministic mitigation based on "physical certainty," binding a process's identity directly to its physical hardware footprint.
7	Standard Instruments Adopted	Windows Native Tools integration (FSRM, PerfMon)	To ensure tool validity and eliminate external software overhead (Minimal Resource Footprint), which could otherwise skew the accuracy of the measured response latency.

Table 3.2: Methodological Framework, Design Decisions, and Technical Rationales.

3.2.8 Quantitative Research Design

This study is based on a practical experimental research design that relies on the establishment of a fully equipped, isolated laboratory. This design is suitable for research aimed at testing algorithms in cyber systems, given the need for precise experimental environments to ensure the validity of the results. This is due to the following methodological reasons:

- **Control over Variables:** Given the need for a controlled experimental environment in this study, the isolated lab allows the researcher to fully control independent variables (such as the type of ransomware attack and encryption speed) and precisely measure their impact on dependent variables (such as response time and resource consumption) without interference from random system processes.
- **Validity of Results:** This design provides a "tested" environment that simulates the technical reality but within a controlled framework, ensuring that the achieved results (such as achieving near-zero latency) are a direct product of the proposed algorithm's logic and not external factors.
- **CyberSecurity Isolation:** Since the study involves dealing with real malicious software samples (such as WannaCry and Locky), the Isolated Lab experimental design is the only methodology that ensures the integrity of external networks and prevents threat leakage during the testing process.

3.3 Data Collection and Measurement Instruments

This section reviews the nature of the data used in the study and how it was collected, in addition to the tools and technical means used to measure the research variables to ensure the accuracy of the results.

3.3.1 Data Sources

The study relies on two main types of data to ensure comprehensive testing:

- **Primary Data:** This is technical data generated in real time within the virtual laboratory environment (Test platform) as a result of ransomware attacks. This data includes system logs and hardware counter readings.
- **Malware Datasets:** Active ransomware samples constitute the "technical study community." To achieve an accurate and comprehensive representation of all current threats, a purposive sample of real-world ransomware targeting Windows-based systems was selected. This sample tested all levels of defense within the hybrid response algorithm by classifying it into two main categories based on encryption and evasion tactics:
  - **Standard Crypto-Ransomware:**
    - Sample examples: WannaCry and Locky.
    - Technical characteristics: Relies on mass and rapid file extension modification and renaming.

* Methodological objective of the selection: To measure the efficiency and response speed of the algorithm's first layer of defense, namely the FSRM-based file screening mechanism and real-time interaction with distributed honeypots.

– **Advanced Stealthy and In-place Ransomware:**
  * Sample examples: Ryuk, Cerber, and custom specialized scripts.
  * Technical Description: These ransomware employs in-place encryption, silently modifying the file's contents without altering its name or extension. This evades traditional monitoring systems and AI/ML probabilistic models.
  * Methodological Objective: To test and challenge the algorithm's "final failure-safe," which aims to establish "physical certainty" by measuring the inevitable, sudden spike in hardware resource consumption (e.g., CPU and Disk I/O) and directly linking it to the process ID (PID).

Methodological Concept (Unit B)	**Empirical Mapping to the Ransomware Framework**
Target Population	Active, real-world Windows-based ransomware threats targeting server environments (Purposive Sampling).
Target Study Community	**Standard Crypto-Ransomware** (WannaCry, Locky) and **Advanced In-place Ransomware** (Ryuk, Cerber, custom scripts).
Unit of Analysis	The full simulated ransomware attack scenario or malicious execution session per virtual testbed instance.
Unit of Observation	Individual time-series telemetry data points: discrete FSRM file screening alerts and hardware counter metrics (CPU and Disk I/O).
Data Source & Trace	Real-time digital traces harvested via system logs, Performance Monitor (PerfMon) updates, and PowerShell runtime counters.

Table 3.3: Methodological Mapping of Computing Units: Analysis vs. Observation for the Ransomware Dataset.

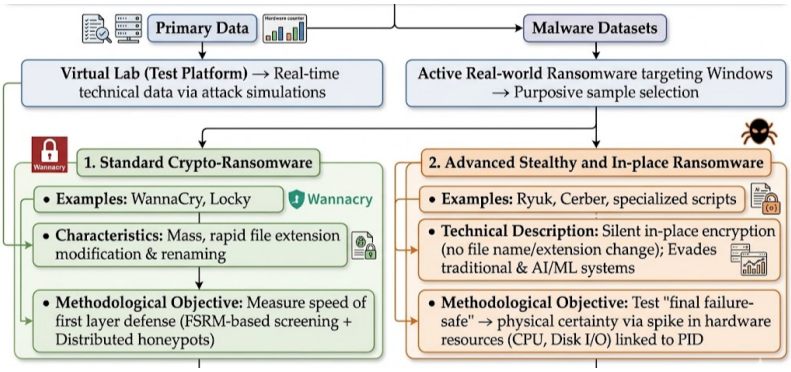


Figure 3.2: Ransomware Study Data Sources.

3.3.2 Technical Measurement Tools

According to the proposed hybrid algorithm, native Windows tools were used as primary measurement tools to reduce the false positive rate and minimize system load and resource usage.

- **FSRM (File Server Resource Manager):** Used as a proactive measurement tool to monitor unauthorized access to honeypots and detect changes in file extensions.
- **PerfMon (Performance Monitor):** Used as a precise physical measurement tool to collect time-series data related to CPU usage and disk I/O read/write operations.
- **Process ID (PID):** Used as a linkage tool to correlate performance readings with the source code of the suspicious activity.
- **PowerShell Runtime Timestamps:** This technical instrument is utilized to measure response latency and automated mitigation execution speeds, quantified precisely in milliseconds (ms)

Readiness Criteria	**Empirical Implementation & Verification**
Session Reconstruction Rules	Events are bounded and linked chronologically using the unique Windows Process ID (PID) combined with the user-session context to reconstruct the exact attack timeline.
Timestamp Integrity	System events and counter polls are stamped in real-time with millisecond (ms) precision via PowerShell runtime counters to guarantee proper alignment between FSRM and PerfMon.
Handling Missing / Null Data	Instances of missing telemetry during hypervisor suspension or severe resource starvation are isolated and excluded from the statistical model to maintain accuracy.
Data Leakage Prevention	The analysis pipeline ensures strict data isolation by operating entirely inside an air-gapped, controlled virtual laboratory environment (Sandbox).
Trace Storage & Formats	Raw metrics are structured and exported into immutable, standardized CSV formats and plaintext event logs for forensic auditability and reproducible evaluation.

Table 3.4: Digital Trace Readiness Card for the Hybrid Telemetry Dataset.

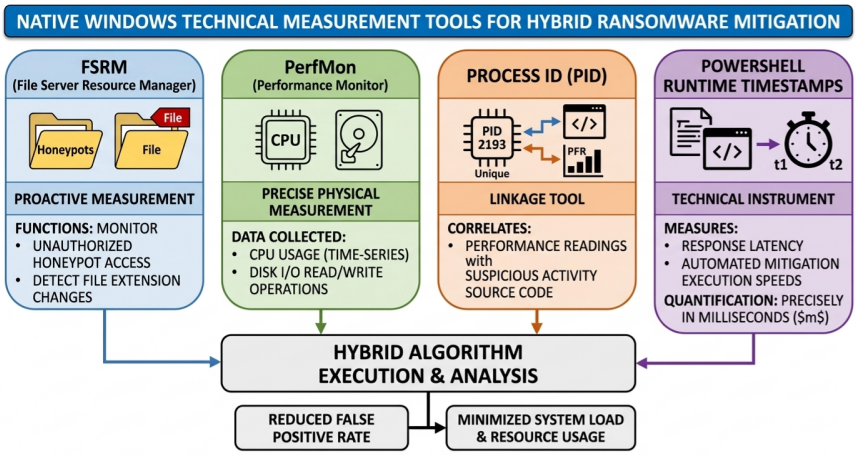


Figure 3.3: Conceptual Technical Measurement Tools

3.3.3 Data Sources and Measurement Instruments

The empirical evaluation of the hybrid ransomware detection framework necessitates a multi-layered data architecture. Figure 3.4 conceptualizes the full structural pipeline, tracking the adversarial stimulation from its initial system baseline disruption down to the generated empirical logs.

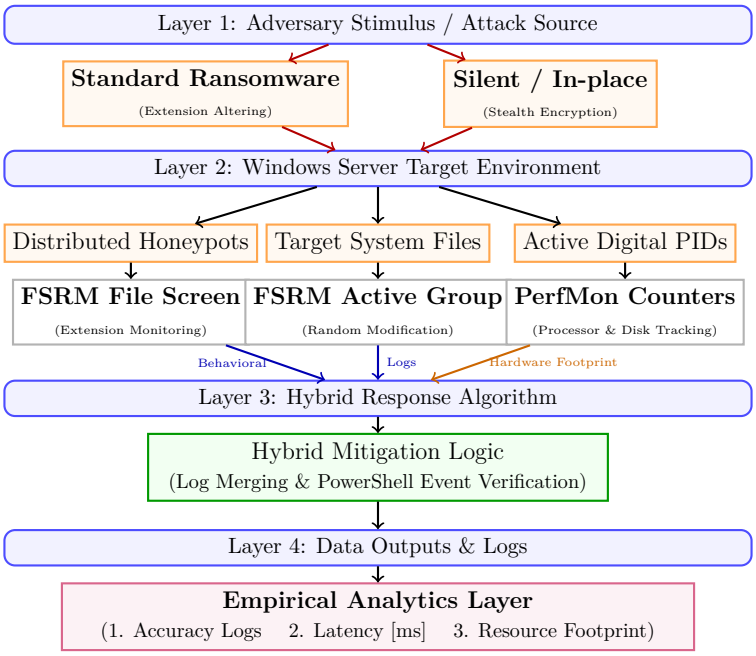


Figure 3.4: Multi-Layered Architectural Flow of the Hybrid Ransomware Detection and Technical Measurement Framework.

3.4 Experimental Environment and Simulation

To achieve the study objectives and test the proposed algorithm, a Virtual Isolated Test platform was created to ensure a realistic simulation of ransomware attacks while maintaining the security of the external network.

3.4.1 Test platform specifications

Virtualization technology was used to build the experimental scenarios. The following table shows the technical specifications of the software and hardware components used in the experiment:

Component	**Specifications**
Physical Host Machine	Laptop with Intel Core i7 Processor, 16GB RAM
Virtualization Software	VMware Workstation Pro
Guest Operating System	Windows Server 2019 / 2022
Virtual Resources Allocation	2 Virtual CPUs (vCPUs), 4GB RAM, 60GB Virtual Disk
Network Configuration	Host-Only (Isolated Network) to prevent malware leakage
Security Tools (Native)	FSRM (File Server Resource Manager), Perfmon (Performance Monitor)
Development Tools	PowerShell (for Algorithm Implementation & Native Tools Integration)

Table 3.5: Experimental Environment and System Specifications

3.5 Environment and Version Freeze Control

To ensure absolute experimental reproducibility, internal validity, and the elimination of external systemic noise, a strict **Environment and Version Freeze Policy** was implemented within the isolated virtual testbed. Since the dependent variables rely on high-precision physical telemetrysuch as CPU percentage and Disk I/O bytes captured via Performance Monitor (PerfMon)any background variance caused by automatic updates or system modifications would compromise the deterministic certainty of the hybrid response algorithm.

Consequently, the software stack, operating system builds, and hypervisor configurations were completely locked (frozen) prior to the execution of the ransomware simulation scenarios (WannaCry, Ryuk, etc.). Table 3.6 details the locked parameters and environmental controls enforced throughout the testing phase.

Component Category	Locked State / Freeze Parameter	Methodological Rationale
Operating System Build	Windows Server 2019/2022 (Stable Build, Automatic Updates Disabled)	Prevents background patch downloads and OS telemetry noise from distorting PerfMon CPU/RAM counters.
Virtualization Layer	VMware Workstation Pro (Host-Only Virtual Network, Isolated)	Guarantees zero network leakage while maintaining identical virtual hardware resource allocation (2 vCPUs, 4GB RAM).
Native Tools Configuration	FSRM Rules, FCI Classifications, and PerfMon Counters	Configured to static baseline thresholds; no dynamic rule additions allowed during live attack injections.
Scripting Environment	PowerShell Runtime Environment (Execution Policy Locked)	Ensures automated mitigation scripts (Stop-Process / FCI tagging) execute under identical execution latencies (ms).

Table 3.6: Environment and Version Freeze Control Matrix for Experimental Reproducibility.

3.6 Simulation Scenarios and Experimental Design

To evaluate the detection mechanisms and response latencies under various attack vectors, several controlled simulation scenarios were executed. Table 3.7 outlines the specific ransomware families and execution parameters used in the experimental testbed to validate the hybrid response algorithm.

Scenario ID	Threat / Attack Vector	Active Instrumentation Stack	Mitigation & Tactical Target
Scenario 1: Stable Baseline	No malware injection. Executing typical enterprise tasks (AD DS queries, local file compression, and backup).	Performance Monitor (PerfMon) idle logs.	Log clean system counters to map baseline thresholds for CPU and Disk I/O.
Scenario 2: Standard Crypto-Ransomware	Rapid, mass file extension renaming and structural path modifications (e.g., WannaCry, Locky variants).	FSRM File Screening Role + Distributed Deception Honeypots.	Block known extension patterns immediately at 'Moment Zero', log the corresponding file-screen violation event (Event ID 8215), and halt the operation.
Scenario 3: Advanced Stealthy Ransomware	Intelligent, slow 'In-place Encryption' modifying file contents without altering extensions or names (e.g., Ryuk, Cerber).	Real-time PowerShell PID analyzer via PerfMon hardware telemetry.	Cross-verify hardware resource anomalies, isolate malicious process ID (PID), and execute immediate kill-switch.
Scenario 4: Multi-Vector Evasion	Simultaneous execution of silent/in-place encryption alongside anti-honeypot evasion scripts.	Full Hybrid Stack (Concurrent FSRM and PerfMon cross-validation).	Validate algorithm resilience against evasion, apply FCI tags ('Under Attack'), and enforce session isolation.

Table 3.7: Simulation Scenarios Matrix and Tactical Native Defense Targets .

Adversarial Dataset and Specimen Characterization

The ransomware strains selected for the simulation grid represent real-world next-generation threats that directly exploit server vulnerabilities:

- **Mass Renaming Threats (WannaCry/Locky):** Deployed to test the signatureless passive-to-active file screening layer of FSRM. These strains target default administrative network shares by appending distinct malicious cryptographic extensions.
- **In-Place Silent Encryptors (Ryuk/Cerber):** Deployed to validate the core physical safety valve of the algorithm. Since these strains deliberately mimic benign system behaviors by evading software-based triggers, their containment relies entirely on PerfMon tracking sustained processor peaks ($> 60\%$) and high-frequency IO data bytes anomalies.

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